

VARSHIL PRAJAPATI

✉ vh.prajapati1215@gmail.com ☎ +91 9327045463
🌐 [LinkedIn](#) 🐙 [GitHub](#) 📍 Pune, Maharashtra, India

Professional Summary

AI/ML Engineer with experience building and deploying end-to-end intelligent systems, spanning deep learning, Retrieval-Augmented Generation (RAG), backend development, and cloud deployment. Published researcher with contributions in AI-driven ECG analysis and smart grid forecasting. Passionate about developing scalable AI applications by combining machine learning, modern LLM frameworks, and backend engineering.

Education

- **M.Tech in Artificial Intelligence & Machine Learning** May 2026
Symbiosis Institute of Technology, Pune
- **B.E in Electrical Engineering** May 2023
L. D. College of Engineering, Amdavad

Technical Skills

Programming Languages: Python, HTML/CSS

ML/AI: PyTorch, PyTorch Lightning, ONNX Runtime, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, LangChain, Pandas, NumPy

Web Frameworks & Tools: FastAPI, React.js, JavaScript

Cloud & DevOps: AWS EC2, Google Cloud Run, Docker, Git/GitHub

Databases: Microsoft SQL Server, Qdrant (Vector Database)

AI Specializations: Natural Language Processing, Generative AI (RAG, LLMs, Prompt Engineering)

Professional Experience

- **Data and Software Engineering (Official Title: Data Analyst Intern)** Aug 2025 – July 2026
Hyster Yale Materials Handling, Pune
 - Built an end-to-end RAG pipeline for financial filings using Python, hierarchical parent-child chunking, Vertex AI embeddings, and Dockerized Qdrant on Google Cloud Run, supporting ~200K child embeddings and 90K parent chunks.
 - Integrated ContextualAI reranking and LangGraph iterative retrieval, achieving 95% hit rate@3 and 0.87 MRR@3 across 500 single-hop queries; evaluated RAG quality with RAGAS, achieving 0.94 faithfulness, 0.98 context precision, and 0.92 context recall.
 - Automated extraction, cleaning, and preprocessing of 150+ enterprise project records using Python/Pandas, reducing manual effort by ~99%; optimized legacy SQL queries by 98-99%, reducing runtime from ~60 minutes to under 5 minutes.
 - Enhanced React.js and SQL applications by delivering new features and resolving production bugs.
- **Technology Intern** Feb 2023 – April 2023
Aatman Corporation, Ahmedabad, India
 - Developed a mobile app for home automation using ETS, iD, and RTI software with KNX protocol integration
 - Assisted in energy audits for two companies by evaluating electrical equipment efficiency

Research & Publications

- **A Comprehensive Review on ECG Diagnosis and Report Generation using AI** *DOI*
Published in Informatics in Medicine Unlocked (Elsevier), 2026
 - Conducted a comprehensive review of **100+ research papers** covering deep learning, transformer architectures, large language models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI for automated ECG diagnosis and clinical report generation

- Analyzed state-of-the-art ECG datasets, evaluation metrics, multimodal learning, and explainable AI techniques, identifying current challenges and future research directions for clinically reliable AI-assisted cardiology systems
- **Smart Grid Load Forecasting**
Published in PACIS 2025 Conference [View Publication](#)
 - Architected ensemble forecasting system combining Time Augmented Transformer, LSTM, and ANN models, achieving 93% R^2 accuracy and 3.63% MAPE on smart grid load prediction datasets
 - Optimized weighted bagging ensemble reducing prediction error to MSE 3.47 and RMSE 1.92, improving forecast reliability by 18% compared to individual models for utility-scale energy management

Projects

Artificial Intelligence & Machine Learning

- **CardioSense – ECG Diagnosis & Clinical Report Generation Platform**
FastAPI, ONNX Runtime, Qdrant, AWS EC2 [Live](#) — [GitHub](#)
 - Developed and deployed an end-to-end AI system for automated 12-lead ECG diagnosis and explainable clinical report generation using Retrieval-Augmented Generation (RAG)
 - Designed a semantic retrieval pipeline using **QdrantCloud**, achieving an average **Precision@5 of 0.83** for clinically relevant report retrieval
 - Optimized deep learning inference using **ONNX Runtime** and deployed a **FastAPI** backend on an **AWS EC2** Linux instance, exposing REST APIs for real-time inference
 - Integrated a Streamlit frontend with the cloud-hosted backend, enabling ECG upload, diagnosis, retrieved case visualization, and AI-assisted clinical interpretation
- **SearchAgent – Full-Stack Agentic Research Platform**
LangGraph, FastAPI, React, Google Cloud Run, Supabase [Live](#) — [GitHub](#)
 - Built and deployed a full-stack AI research platform with an 8-node cyclic LangGraph and FastAPI backend on Google Cloud Run and a React 19 and Vite frontend on Vercel, supporting web search, Wikipedia retrieval, sandboxed Python execution, and Groq LLM generation
 - Implemented production reliability with XML output contracts, loop guards, context truncation, backup generation, AST-validated process-isolated code execution, Pydantic APIs, MemorySaver state, and Arize Phoenix tracing
 - Created a real-time WebSocket interface with streamed agent status and responses, Markdown, table, code, and image rendering, Supabase authentication and conversation persistence, guest access, and resilient loading and error states

Data Science & Analytics

- **JSON ML Pipeline**
Machine Learning, Feature Engineering, XGBoost [GitHub](#)
 - Built automated end-to-end ML pipeline processing complex nested JSON structures, reduced feature engineering time by 70% through automated extraction and flattening
 - Trained and optimized Random Forest and XGBoost models with hyperparameter tuning, achieving 88% predictive accuracy on multi-class classification task

Certifications

- **LangChain for LLM Application Development** – DeepLearning.AI – [DeepLearning.AI](#)
- **Data Science Specialization** – [AlmaBetter](#)
- **Google Cloud Essentials** – [Google Cloud](#)
- **Data Visualization Tools & Business Analysis Fundamentals** – [AlmaBetter](#)

Achievements

- **GATE 2024 Qualified** – Data Science & Artificial Intelligence (DA) with score of 353, top 15 percentile
- **HackerRank 5-Star Gold Badges** – SQL and Python Programming, demonstrating advanced algorithmic problem-solving skills